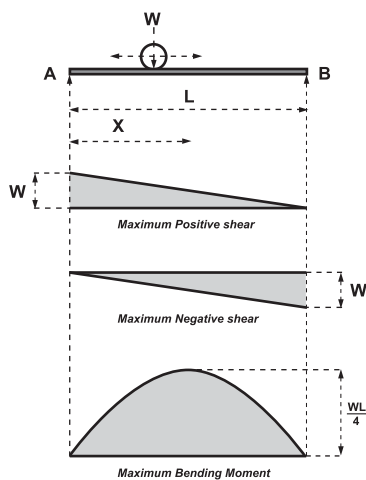


Simply supported beam



Single concentrated moving load

Maximum Positive Shear at any section occurs when the load is immediately to the right of the section. Similarly, Maximum Negative Shear occurs when the load is to the left. For a section distance X from A:

$$\text{Positive } F_{x \max} = W \frac{L-X}{L} \quad \text{Negative } F_{x \max} = -W \frac{X}{L}$$

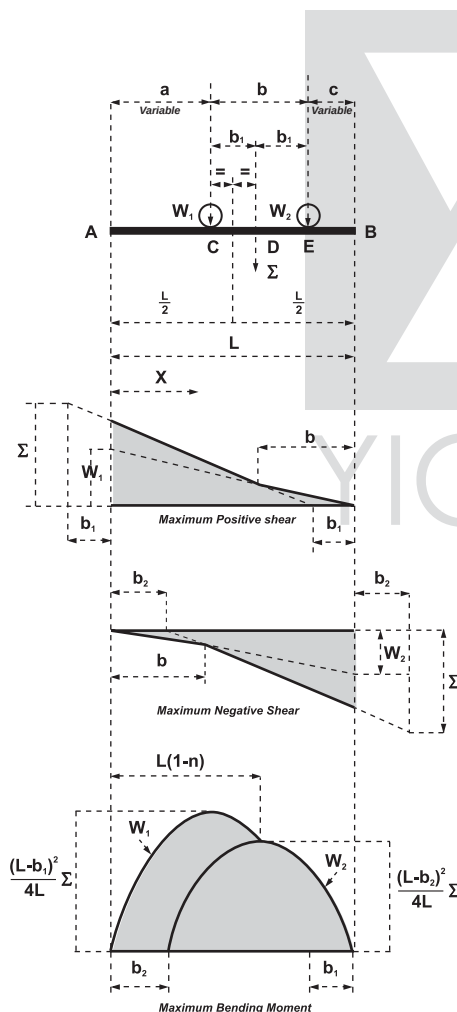
Maximum Bending Moment at any section occurs when the load is over the section. For a section distance X from A:

$$M_{x \max} = W \frac{X(L-X)}{L}$$

The Absolute Maximum Bending Moment and Deflection occur under the load at mid-span

$$\left. \begin{array}{l} M_{\max \max} = \frac{WL}{4} \\ \delta_{\max \max} = \frac{WL^3}{48EI} \end{array} \right\} \quad \left. \begin{array}{l} \text{Maximum end slope at A occurs} \\ \text{with the load at } X=0.42265L \\ \text{from A.} \end{array} \right\} i_{A \max} = 0.06415 \frac{WL^2}{EI}$$

Simply supported beam



Two concentrated moving load

Assume : $W_1 > W_2$ $W_1 + W_2 = \Sigma$ $W_2 = n\Sigma$
Fixed Distance $b = mL$

$$b_1 = \frac{W_2 b}{W_1 + W_2} = nmL \quad b_2 = (m-n)L$$

Maximum Reaction at A and Absolute Maximum Positive Shear occur when W_1 is immediately to the right of A:

$$R_{A \max} = \text{Positive } F_{\max \max} = W_1 + W_2 \quad \frac{L-b}{L}$$

For a section distance X from A:

$$X \leq L-b \quad : \quad L-b \leq X$$

$$\text{Positive } F_{\max} = \frac{L-X}{L} m W_2 \quad : \quad \text{Positive } F_{\max} = \frac{L-X}{L} W_1$$

Note: 1. For $R_{B \max}$, interchange values of W_1 and W_2 in the formula for $R_{A \max}$

2. For Negative Shear, interchange W_1 and W_2 in formulae for Positive Shear, measuring X from B towards A

3. If $m > \frac{n}{1-n}$ calculate $R_{B \max}$ and Negative Shear values for W_1 only as single load.

Absolute Maximum Bending Moment occurs under W_1 when that load and the resultant of both loads are equidistant from mid-span (see loading diagram):

$$M_{\max \max} = \frac{(L-b_1)^2}{4L} \Sigma$$

If $m < n$, the Maximum Bending Moment at any section occurs under one of the loads. For a section distance X from A:

$$X \leq L(1-n)$$

$$M_{\max} \text{ under } W_1 = \frac{(L-b_1-X)X}{L}$$

$$L(1-n) \leq X$$

$$M_{\max} \text{ under } W_2 = \frac{(X-b_2)(L-X)}{L}$$